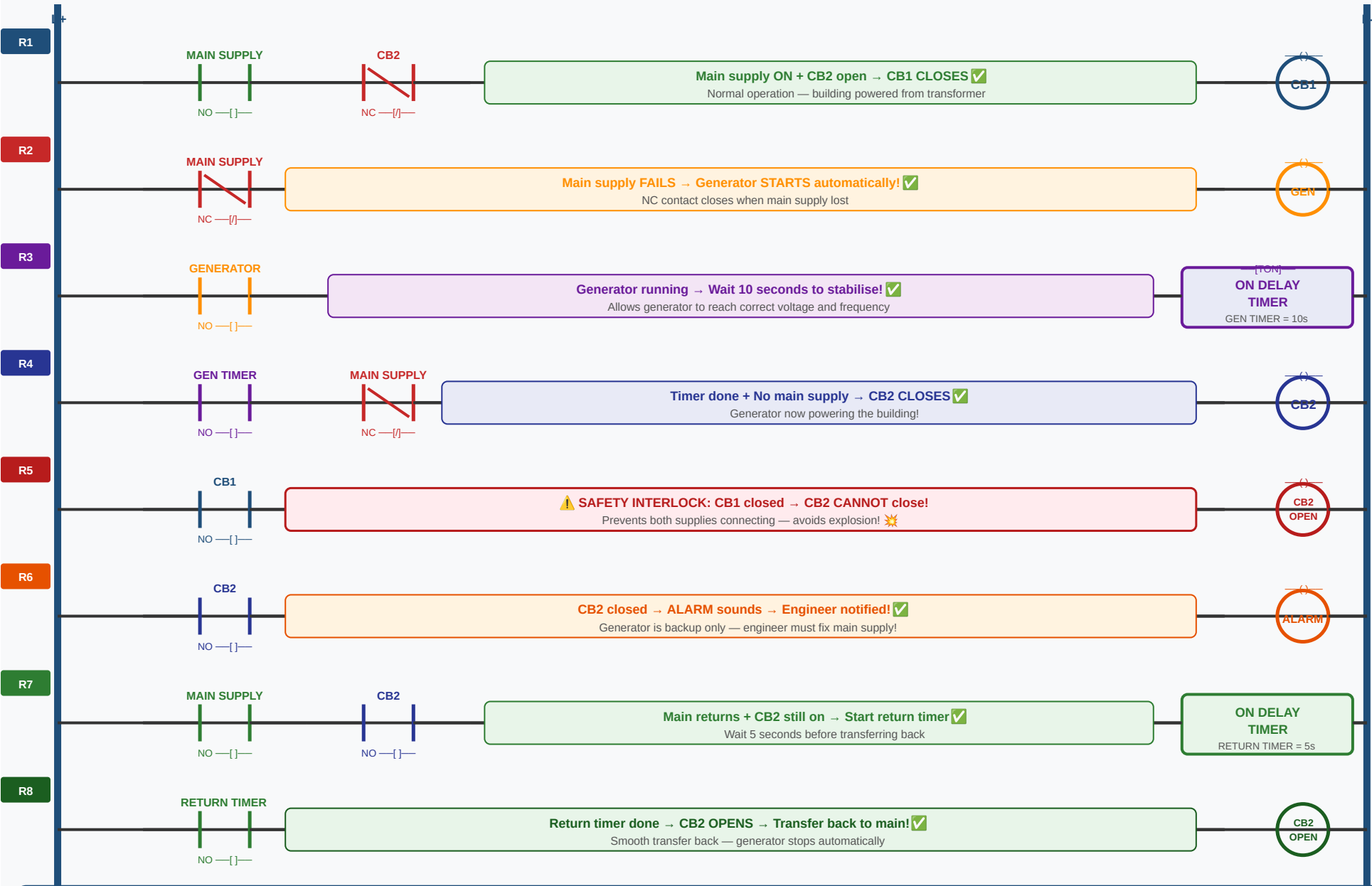


Automatic Transfer Switch — ATS Ladder Logic

Mohammed Azam Ali MIET — Schneider Electric — N+1 Redundancy System



COMPLETE OPERATING SEQUENCE — N+1 REDUNDANCY

NORMAL OPERATION

- ✓ Main Supply ON
- ✓ CB1 CLOSED
- ✓ CB2 OPEN
- ✓ Generator OFF
- ✓ Building powered from transformer

MAIN SUPPLY FAILS

- ✗ Main Supply FAILS
- ✓ CB1 OPENS (R1)
- ✓ Generator STARTS (R2)
- ✓ Timer counts 10s (R3)
- ✓ CB2 CLOSING (R4)
- ✓ ALARM sounds (R6)

GENERATOR RUNNING

- ✓ Generator powers building
- ✓ CB2 CLOSED
- ✓ CB1 OPEN
- ✓ Alarm ON
- ⚠ Engineer investigates main supply fault

MAIN SUPPLY RETURNS

- ✓ Main supply restored
- ✓ Return timer 5s (R7)
- ✓ CB2 OPENS (R8)
- ✓ CB1 CLOSING (R1)
- ✓ Generator STOPS
- ✓ Back to normal! ✓